

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Pharm. Examination

University Theory Examination December 2017

PH419 Pharmaceutical Technology – II

Date: 04.12.2017 , Monday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. _____ is a thermodynamically stable formulation.
[A] Microemulsion
[B] Emulsion
[C] Multiple emulsion
[D] Suspension
 2. For aerosol formulation cascade impactor is used for _____.
[A] Determining particle size
[B] Identification of propellant
[C] Flash point
[D] Determining moisture content
 3. Peach like effect is not given by _____ ingredient of face powder.
[A] Rice Starch
[B] ZnO
[C] Maize Starch
[D] Powdered Silk
 4. Which of the following is not a hair remover?
[A] Zinc sulphide
[B] Copper sulphate
[C] Calcium sulphide
[D] Arsenic trisulphide
 5. Particle size of an emulsion cannot be measured using _____.
[A] Microscopic method
[B] Coulter counter
[C] Brookfield viscometer
[D] Laser Light scattering method

6. Addition of _____ to nail polish preparation improves flexibility and lasting power.
[A] Di-octyl phthalate
[B] Di-butyl stearate
[C] Acetylated monoglycerides
[D] Resins
7. Space sprays delivers particles up to _____ micron.
[A] 20
[B] 6
[C] 15
[D] 50
8. Which mechanical stirrer is used for low viscous emulsions?
[A] Paddle
[B] Turbine
[C] Anchor paddle
[D] Propeller
9. Miscibility test is used for identification of _____.
[A] Suspension
[B] Emulsion
[C] Solution
[D] Syrup
10. Butylated hydroxyl toluene (BHT) is used as _____ in sterile products.
[A] Wetting agent
[B] Antimicrobial preservative
[C] Anti-oxidants
[D] Suspending agent
11. Tag open cup apparatus is used to measure _____ of aerosols.
[A] Vapor pressure
[B] Density
[C] Flame projection
[D] Flash point
12. The _____ plays an important role in technology evaluation, scale-up, and transfer activities of new products.
[A] Pilot Plant
[B] Validation
[C] NDA
[D] QA/QC
13. LAL test is positive if _____.
[A] Coagulation with loose gel
[B] No coagulation
[C] Coagulation with firm gel
[D] No gel formation

14. Which of the following wax is not used in lipstick preparation?
[A] White beeswax
[B] Candelilla wax
[C] Ozokerite wax
[D] Cetrimide emulsifying wax
15. Pyrogen can be detected in parenterals by _____.
[A] Rise in body temperature
[B] Decrease in body temperature
[C] No change in body temperature
[D] Rashes on skin
16. Leak test tank is heated upto ____ degree Fahrenheit for testing aerosols.
[A] 103
[B] 130
[C] 131
[D] 113
17. Disposable syringe is made from _____.
[A] Polypropylene
[B] Glass
[C] Transparent polypropylene
[D] PTFE
18. DOP test is usually used for _____.
[A] Filter validation
[B] Filter leak testing
[C] Filter compatibility testing
[D] HEPA filter integrity
19. _____ provides the seal between the ferrule and the aerosol container.
[A] Gasket
[B] Stem
[C] Spring
[D] Dip tube
20. Evaluation of formulation on excised animal skin is _____ type of study.
[A] *In-Vivo*
[B] *Ex-Vivo*
[C] *In-Vitro*
[D] Cell Line study

SECTION – II

- Q 2** Attempt any **TWO** of the following
- A** Enlist vehicles used for parenteral dosage form. Discuss preparation and storage of WFI. 05
 - B** Explain formulation and evaluation of toothpaste in detail. 05
 - C** Enumerate evaluating parameters of aerosols and explain in brief performance tests of the same. 05
- Q 3** Attempt any **TWO** of the following
- A** Distinguish between vanishing cream and cold cream with examples. 05
 - B** Comment: Large volume Parenterals are given by IM route. Describe various types of IV sets. 05
 - C** Comment: Sedimentation rate of flocculated suspensions is faster than deflocculated suspensions. Explain formulation of flocculated suspension. 05

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Discuss in brief about prefilled syringes. 05
 - B** Explain in detail evaluation parameters of loose powders and compact powders. 05
 - C** What do pilot –plant and scale- up mean? State objectives of scale-up. Explain the steps in scale-up. 05
 - D** Describe various types of stabilizers used in liquid dosage form. 05
 - E** Classify routes of drug penetration through skin. Describe various factors affecting on drug penetration through skin. 05
 - F** Discuss the evaluation parameters of Suspensions & Emulsions. 05
- Q 5** Attempt any **FOUR** of the following
- A** Write a note on FFS Technology. 05
 - B** Differentiate creaming and cracking of emulsion. Classify types of emulsifying agents with examples. 05
 - C** Describe the liquified gas and compressed gas systems of aerosols in brief. 05
 - D** Write a note on: Ocusert. 05
 - E** What are clean rooms? Discuss the standards for clean rooms. Enumerate criteria for environmental monitoring of clean rooms. 05
 - F** State the comparative advantages of cold filling & pressure filling techniques in manufacturing of aerosols with a well labeled diagram. 05
-